

A parallel guaranteed projector-camera system for dual videography

Yushi Yamagiwa

Tokai University

Tokyo, Japan

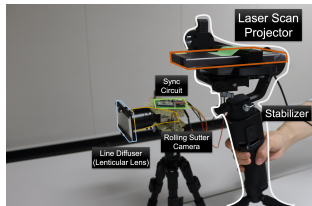
8bj2111@cc.u-tokai.ac.jp

HiroYuki Kubo

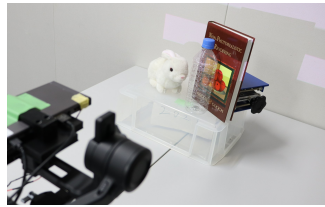
Tokai University

Tokyo, Japan

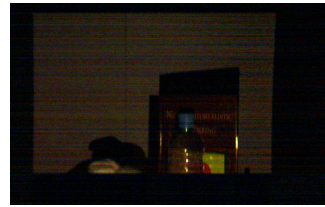
hkubo@tsc.u-tokai.ac.jp



(a) our system



(b) scene



(c) Result:Non-tilting projector



(d) Result:projector tilted by 10°

Figure 1: Our optical system (a). The timing of the raster scanning projector illumination and the rolling shutter camera exposure are synchronized by the circuit. The projector is automatically leveled using a stabilizer. The positional relationship of each object in the scene (b). In contrast to the projector direction in (c), the projector is tilted in (d). As a result, we can obtain a lower view in (d) without changing the camera directions.

ABSTRACT

Thanks to the optical duality between a projector and a camera, an image from the viewpoint of the projector can be computationally generated using camera images; however, to achieve dual photography in real time, the geometric configuration of the projector and the camera must be strongly constrained. In this research, to substantially relax this constraint, we improved the optical components of the projector-camera system by employing a stabilizer to guarantee that both epipolar planes are parallel. As a result, it is possible to move the projector to change the synthesized virtual viewpoint more freely in real-time.

CCS CONCEPTS

• Computing methodologies → Computational photography.

KEYWORDS

dual videography, computational photography, stabilization

ACM Reference Format:

Yushi Yamagiwa and HiroYuki Kubo. 2021. A parallel guaranteed projector-camera system for dual videography. In *SIGGRAPH Asia 2021 Posters (SA '21)*, December 14–17, 2021, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3476124.3488621>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA '21, December 14–17, 2021, Tokyo, Japan

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8687-6/21/12.

<https://doi.org/10.1145/3476124.3488621>

1 INTRODUCTION

An image from the viewpoint of a projector that is computationally generated using image(s) from a camera, is called dual-photography [Sen et al. 2005]. Using Helmholtz's conflict theorem, Sen et al. [2005] succeeded in acquiring the image of the projector's viewpoint from the light incident from the projector to the sensor. However, this method computationally obtains the image from the projector's viewpoint using captured images. Because many image captures are required, it is difficult to acquire dual-photography images in real time. Therefore, their method is not very suitable for handling continuously captured images. O'Toole et al. [2015] proposed dual videography which is a method to acquire images from the projector's viewpoint in real time using a time-synchronized projector-camera system. However, this method requires the camera and projector to be parallel because the captured image is distorted when the projector rotates around the irradiation axis. In addition, the contrast of the generated images tends to be low, because the optical system of dual videography [O'Toole et al. 2015] uses a line diffuser in front of the camera. Therefore, we propose an optical approach to solve this problem. In other words, by automatically stabilizing the relative positions of the projector, we can relax the positional constraints of the continuous-capture method for images from the projector's viewpoint. Moreover, by taking the difference between the captured image and the ambient light image, we can remove the ambient light to achieve a better signal-noise ratio for the projector's viewpoint image.

2 METHODS

2.1 AUTO STABILIZATION

Our optical system is shown in Fig. 2 and basically follows the synchronized projector-camera system [O'Toole et al. 2015] which consists of a laser scanning pico-projector and a rolling shutter camera. A line diffuser is installed in front of the camera lens so

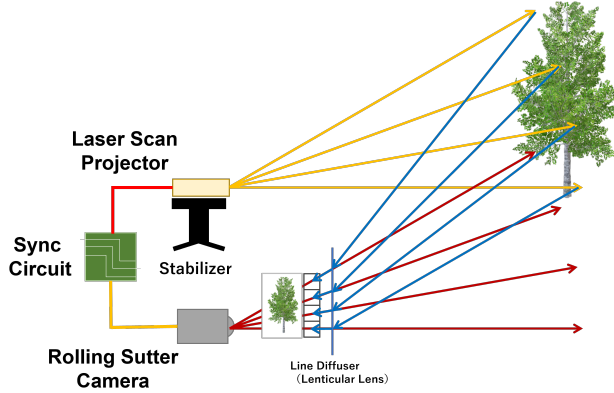


Figure 2: Schematic diagram of our method. The emitted light is focused by a line diffuser and captured by a camera.

that each pixel of the camera has a larger vertical field of view. Because the timing of the projector illumination and exposure of the camera are synchronized, an image from the viewpoint of the projector is simultaneously obtained [O’Toole et al. 2015].

The problem here is that the epipolar planes of the camera and projector must maintain the same orientation. In this research, we built a new imaging system to solve this issue by additionally implementing a stabilizer for the projector to keep the projector plane horizontal. As a result, this automatically stabilizes the epipolar plane of the projector and relaxes the constraint.

2.2 AMBIENT LIGHT REMOVAL

In the dual videography of O’Toole et al [2015], the captured image can be represented as follows: A line diffuser is placed in front of the camera to increase the angle of view in the vertical direction. Therefore, the entire pixel of the captured image is strongly affected by the ambient light and the contrast is reduced. Thus, we apply the following process to the captured image.

$$I = I_c - I_a \quad (1)$$

where I_c represents the captured image and I_a represents the image taken with ambient light only. The ambient-light image is captured using the light from the projector without projection. As a result, the ambient-light image is removed from the captured image, and an image in which the projector is the only light source can be obtained.

3 RESULTS

Fig. 3 presents the result of our automatic stabilization. Fig. 3(a) shows the image when the projector is rotated and Fig. 3(b) is the image obtained using our method. We can see that the image in Fig. 3(a) is highly distorted when compared with the one taken with our method, and it is important to keep both epipolar planes horizontal in this imaging system. In our research, we expect the projector to be carried like a camera. Using our system, the posture of the projector is guaranteed by the stabilizer. Therefore, it is possible to move the projector while perfectly preserving the image.

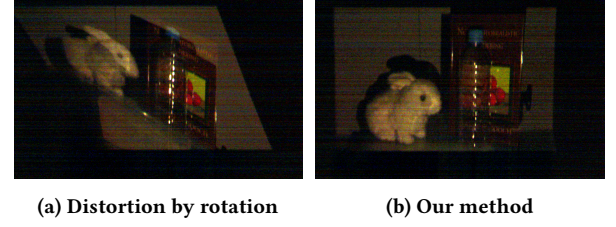


Figure 3: Result of auto stabilize

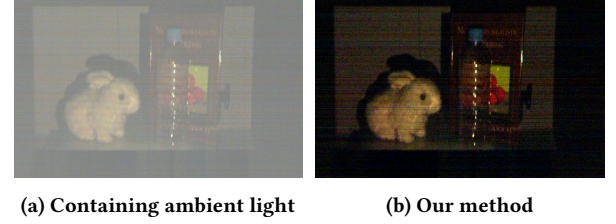


Figure 4: Result of ambient light removal

Fig. 4 is another comparison of the result of removing the ambient light. Fig. 4(a) is the original captured image, and Fig. 4(b) is the image obtained using this method. In the image obtained by our method, the values are multiplied by a factor of ten, whereas in the captured image, they are multiplied by four. As a result, the entire image in Fig. 4(a) is whitish. This is because the captured image contains ambient light. By contrast, our method removes the ambient light, so only the part of the image illuminated by the projector is extracted. This results in a more complete image with higher contrast. We used DJI’s Ronin-SC stabilizer for this experiment.

4 CONCLUSION

In this study, we realized a method for capturing images from the projector’s viewpoint in real-time without distortion caused by mis-alignment between a projector and camera. More specifically, we relaxed the constraint of the positional relationship between the projector and camera using an optical approach based on a stabilizer. Furthermore, we reduced unnecessary ambient light in the environment using an image-processing approach. As a result, we have improved the real-time acquisition of dual videography. As future work, we will extend this optical system to implement a new imaging method.

ACKNOWLEDGMENTS

This work was supported by JSPS Kakenhi(JP19H04138, JP21K19799), JST CREST (JPMJCR1764), and JST FOREST(JPMJFR206I). We thank Kimberly Moravec, PhD, from Edanz(<https://jp.edanz.com/ac>) for editing a draft of this manuscript.

REFERENCES

- Matthew O’Toole, Supreeth Achar, Srinivasa G. Narasimhan, and Kiriakos N. Kutulakos. 2015. Homogeneous Codes for Energy-Efficient Illumination and Imaging. *ACM Trans. Graph.* 34, 4, Article 35 (July 2015), 13 pages. <https://doi.org/10.1145/2766897>
- Pradeep Sen, Billy Chen, Gaurav Garg, Stephen R. Marschner, Mark Horowitz, Marc Levoy, and Hendrik P. A. Lensch. 2005. Dual Photography. *ACM Trans. Graph.* 24, 3 (July 2005), 745–755. <https://doi.org/10.1145/1073204.1073257>